

# The Digital Landscape of Bluesky: Structure, Influence, and Connectivity

## Analysis of Yanis Varoufakis' Network

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### Abstract

*This coursework examines the Bluesky network of Yanis Varoufakis, a prominent political figure of MeRA25/DiEM25, to investigate engagement patterns and influential users. Using Gephi and NetworkX, the study analyzes the network's structural properties, including node and edge counts, network diameter, average path length, density, and centrality measures. The findings reveal a fragmented network with low clustering, a high number of weakly connected components, and a significant presence of isolated nodes. Various centrality measures highlight differing sources of influence within the network, with Varoufakis ranking highest in PageRank but not in closeness or eigenvector centrality. These insights provide a preliminary understanding of Bluesky's evolving social dynamics and its divergence from traditional centralized platforms.*

## 1 Introduction

Social media has played a fundamental role in shaping digital communication and on-line interaction for several decades. From the early days of instant messaging services such as MSN Messenger and AOL Instant Messenger in the late 1990s to the rise of social networking platforms like Facebook and Twitter, the landscape of online social platforms has continuously evolved. In recent years, short-form video content has come to dominate the social media market, with platforms such as TikTok. On the other hand, Bluesky is the next era of social media; it builds on the short text format that Twitter popularized but introduces the decentralized nature that other cutting-edge technologies also use (for example Bitcoin with its distributed ledger).

Bluesky builds upon the short text-based format that was originally popularized by Twitter but introduces a decentralized architecture that aligns with broader trends in

digital innovation. This decentralized approach allows users greater autonomy over their digital presence. Furthermore, Bluesky embraces an open-source framework, granting developers transparency and the ability to contribute to or modify the platform's underlying code. Perhaps most significantly, Bluesky enables individuals to host their social media profiles on their own infrastructure, reducing dependence on centralized corporations and fostering a greater sense of digital sovereignty. These characteristics mark Bluesky as a significant departure from traditional social media platforms, which often rely on centralized control and opaque algorithmic moderation.

Bluesky rose to even larger prominence after the US election and the victory of President Trump. A pivotal moment in Bluesky's expansion came because of actions taken by Elon Musk, who, in addition to being, a Trump appointed, Special Government Employee at the Department of Government Efficiency, is also the majority owner

of X/Twitter, Bluesky's primary competitor. Musk's reforms to Twitter's content moderation policies, particularly those concerning misinformation and the incident in which Musk was filmed performing a "Roman Salute," a gesture historically associated with fascist symbolism [1] led to widespread debate and prompted many users to seek alternative platforms such as BlueSky. Both these actions have been heavily controversial [2] and have prompted legal action from critics [3].

## 1.1 Significance of Network Analysis in Understanding Social Platforms

As extensively discussed in our lectures, Network Analysis plays a crucial role in examining Social Networks and therefore is a strong tool for also analyzing social media platforms. It is particularly helpful in understanding the ways individuals connect and interact. By mapping the relationships between users, analyzing patterns of communication, and identifying influential nodes within these networks, Network Analysis provides valuable insights into the structure and dynamics of digital social ecosystems. This analysis is very relevant in studying emerging platforms such as Bluesky, where the decentralized architecture and the novel governance model redefine traditional online interactions.

## 2 Choosing a Profile

For this coursework, I have chosen to analyze the Bluesky network of Yanis Varoufakis, focusing on the structural characteristics and dynamics of his community. Specifically, I examined key network metrics to identify influential accounts and uncover underlying patterns of interaction. Measures of centrality were employed to determine the

most prominent users within the network. Additionally, clustering coefficients, triadic closures, modularity and homophily were also analyzed. Finally, PageRank was also utilized to further evaluate user prominence in the network.

The main objective of this coursework is highlighting Varoufakis' role in fostering political discourse online and the structural dynamics of his digital community. By analyzing the topology of his connections and the flow of information, this coursework aims to provide interesting insights into his network.

### 2.1 Network Interest

As instated above he is quite active online, I was expecting his network to be quite complex and interconnected, as his views (in my opinion) usually push for the creation of echo chambers. Although after pulling his network, and making initial inferences, I was surprised to be proven wrong. His network was very open-ended and lightly interconnected. In the analysis that follows it is obvious that this network is the opposite of what someone would expect.

### 2.2 Personal Interest

Yanis Varoufakis is a published economist and former Minister of Finance appointed by former Prime Minister Alexis Tsipras during the height of the Greek (and European) debt crisis. His outlook on Economics could be characterized as somewhat controversial [4], something he even seems to agree on [5]. Other than that he is the only Greek politician I could find with a BlueSky account and given my interest in Greek politics I decided I would pick something in that realm. Let us also not forget his extensive work on Economics and Social Policy [6, 7, 8] which makes me believe that his account is worth analyzing.

Please keep in mind I personally neither necessarily agree neither endorse his figure or his views.

### 3 Limitations

It must be made abundantly clear that this analysis comes with some very strong limitations. The preliminary nature of BlueSky's network, the fact that the Gephi plugin seems to not fetch 100% of the data you'd expect, combined with the lack of the ability to completely verify the pulled network's structure creates a cocktail of problems that could be proven fatal for this coursework. Putting that to one side, this does not mean the coursework lacks value; it still has the potential to surface interesting and meaningful patterns within the network. However, it is crucial to recognize that any conclusions drawn from this coursework should be approached with caution rather than them being treated as definitive, they should also be seen as preliminary insights that require further investigation and validation.

#### 3.1 Katz Centrality

Eigenvector centrality led to a big disparity between the first and second accounts, I elected to also try Katz centrality as it extends eigenvector centrality by introducing an attenuation factor that accounts for both direct and indirect connections, ensuring that influence diminishes with distance. This modification allows Katz centrality to prevent dominance of highly connected nodes, making it more versatile for analyzing real-world networks. Even though I implemented the code; it required 51GB of RAM to run, I only have 40GB so it was never computed.

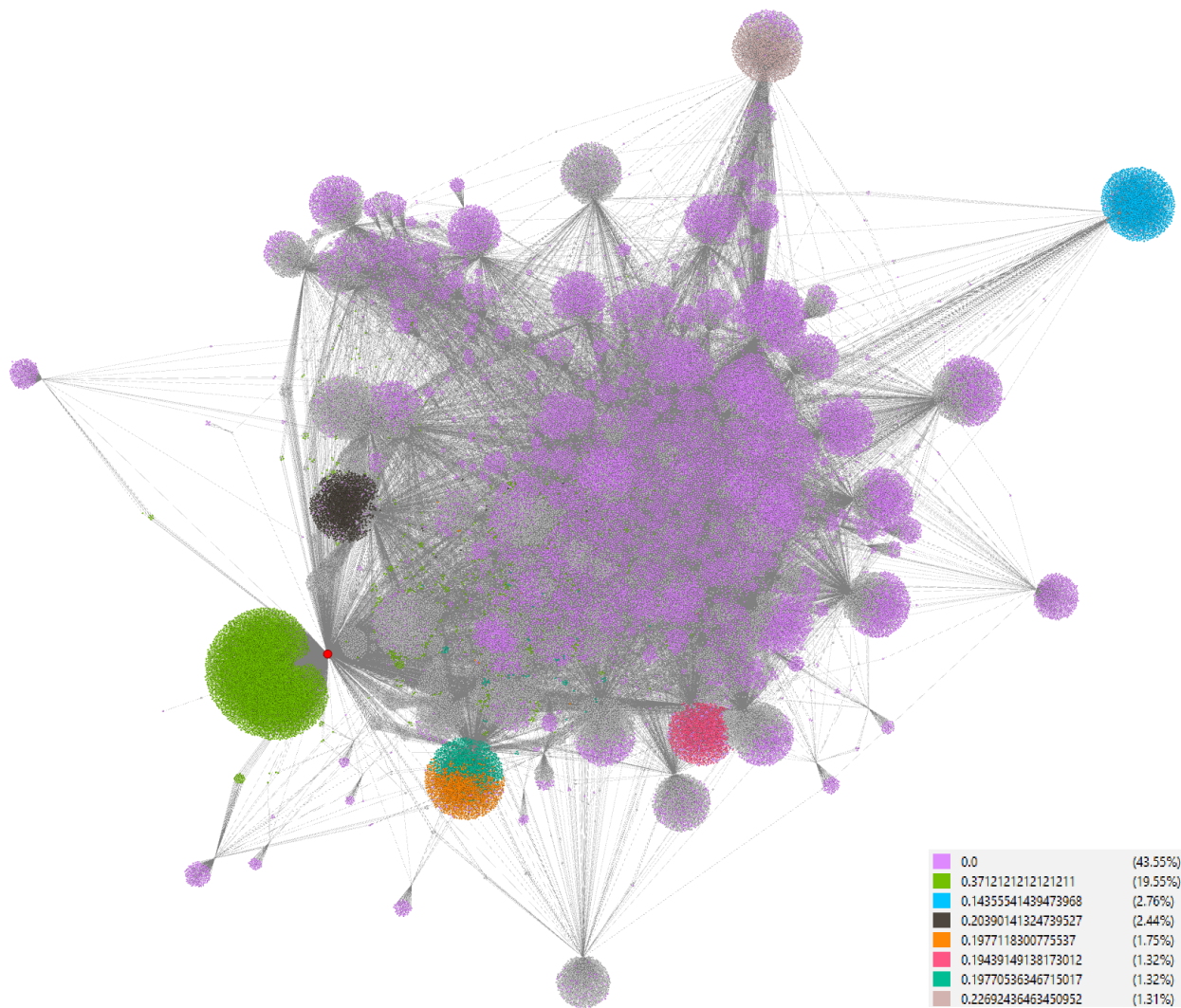
## 4 Methodology

To analyze Varoufakis network, I mainly used Gephi and the Bluesky Plugin available for it. I pulled the data from Bluesky using the plugin and ran most of the analysis using Gephi's included features. For more complex analysis, that could not be performed within Gephi, I used python with the NetworkX library. The code used is available on GitHub here []. Some of the visualizations created by Gephi were, to put it lightly, appalling; so Matplotlib or Seaborn were used to redo them.

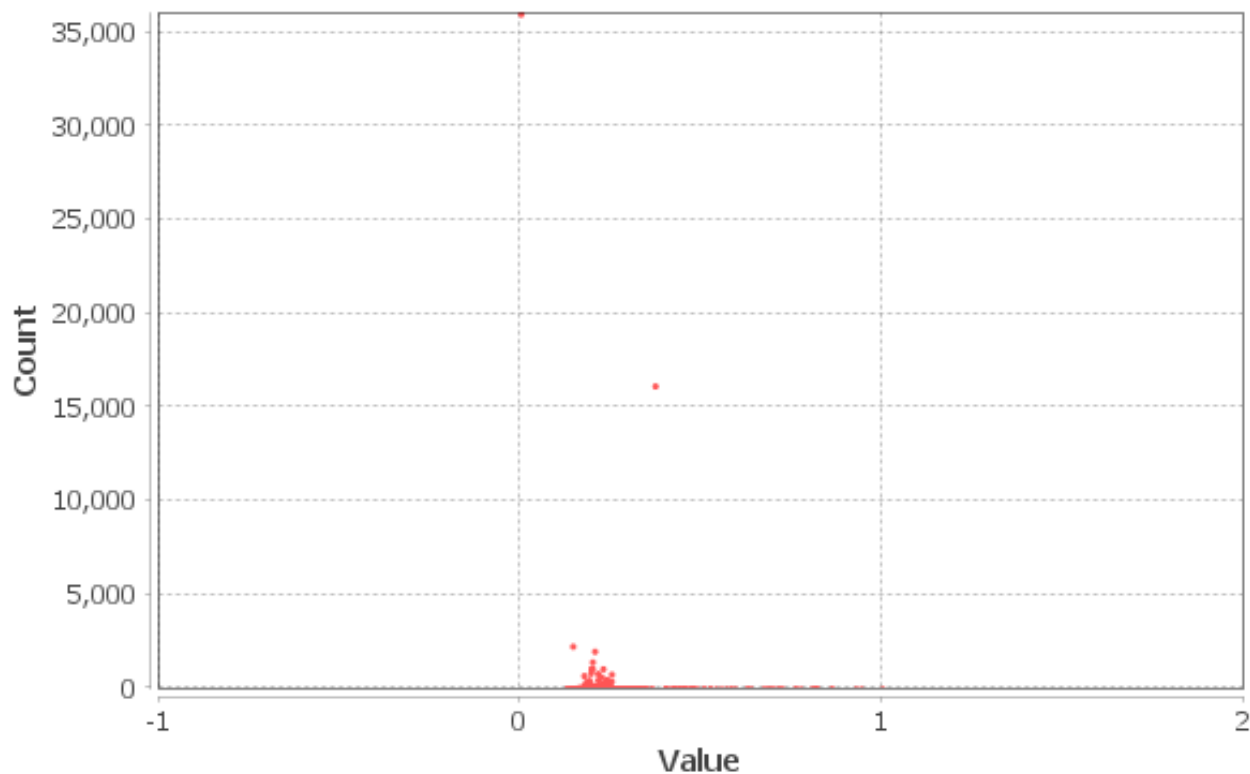
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## 5 Graphical representation of the network

The graphical representation of the network is seen in figure 1. (The nodes have been colored in accordance with the Harmonic Closeness Centrality). You would expect that Gianis was in the center of the vastly interconnected part on the right, but he is actually the larger red node on the bottom left. He seems to be connected strongly with the large partition on the right and with a single link with each node in the partition on the left. He is therefore the only link between these two "network parts". This intrigued me to also investigate what the difference between these two parts is.



**Figure 1:** The Graph colored in respect to the Harmonic Closeness of nodes



**Figure 2:** *The Harmonic Closeness Centrality Distribution*

## 6 Basic topological properties

The network has the following topological properties. First and foremost, the network is directed as edges point from the follower to the followee. It has 82.733 nodes and 133.115 edges. The network diameter is twelve and the average path length is 5.23. Our network could be characterized as being quite large, this is expected as Gianis's followers on BlueSky are, at the time of pulling the nodes, 18.210 the graphs diameter is considered medium sized and the average path lengths are short.

## 7 Component Measures

Component measures help us understand how connected a network is by identifying distinct sub-networks (components) within the larger structure. These measures are

particularly important in analyzing decentralized social platforms like Bluesky, where user connectivity patterns can differ significantly from those seen in centralized social media networks.

There is one Weakly Connected Component in Giannis's network, which is expected as all followers and followees must be connected to him or other nodes in the network in order to be pulled by the plugin. This component is not characterized as being giant, I compare it further later to a Erdős-Rényi random graph and this becomes even more evident.

If we also look at the graph directionally then there is a high number of Strongly Connected Components, this is partly because Gianis doesn't follow many people on Bluesky, therefore most of his followers form a Strong Component of their own. It's even more evident as there are only 13 Strongly Connected Components which contain more than one node in the network but a staggering 68.212 Strongly Connected Components

overall.

As stated earlier, it's easily distinguishable that there is quite a large node area on the left of Gianis, this area seems to contain all his followers that are not connected to anyone else in the graph. This is in line with the color of the degree of these nodes signifying a degree of one (that being the connection to Gianis). Looking at these nodes there were about 6.000 of them that did not contain a Bio (Description). This caught my attention. After digging a bit further and shifting through many of them I found that they rarely contained any self-authored posts, and they were either constantly reposting other posts or their wall was empty. They also, after coming back to them a couple months after pulling our network, are following a small amount of people which seem to be of common interest. I don't want to draw any conclusion, so the inference is left as practice to the reader of this coursework.

## 8 Degree Measures

Degree-based measures provide fundamental insights into the structure of a network by quantifying how well-connected individual nodes are.

In figure 3 we can see how the graph is colored in, in accordance to the Degree of the nodes. An overwhelming number of nodes have a degree of one, which might not be that interesting for the analysis, therefore in figure 4 has a filter applied to only show nodes with a degree greater than one. Notice the complete abolishment of subsection number two (prev. blue). The average degree is 1.609 which is quite low.

As we have thoroughly discussed in class, random networks don't model social networks gracefully, but they are helpful for comparison to draw better conclusions. Bluesky seems to be so new as a social media

app that most accounts are only following a small number of related people. If we compare it to a Erdős–Rényi random network, as described in [9], with an equivalent 82.733 nodes and using the equation

$$\langle k \rangle = \frac{2m}{n}$$

we find that for the same average degree of 1,609 the probability of two vertices to be connected is equal to

$$p = 2 \times 10^{-5}$$

which is incredibly small. This suggests, at least initially, that Bluesky does not resemble a well-connected social network but instead appears highly fragmented, with most users following only a small number of closely related accounts.

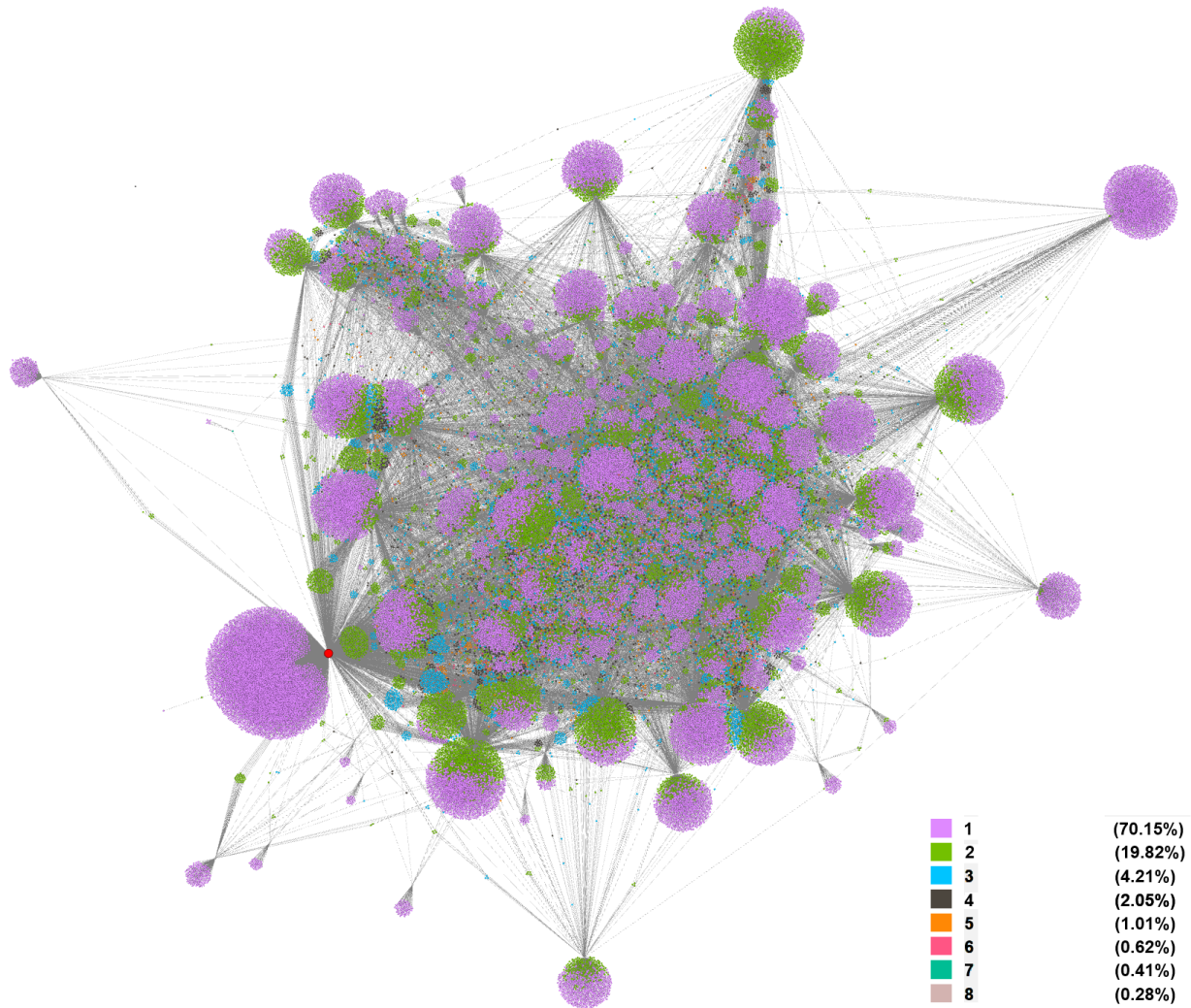
The maximum degree is, as expected, Gianis's node with a degree of 18.211. In figure 5 we can see the distribution of the Degrees, note the very high amount of nodes with a degree of one and how quantity of nodes drops off for higher degrees.

## 9 Centrality Measures

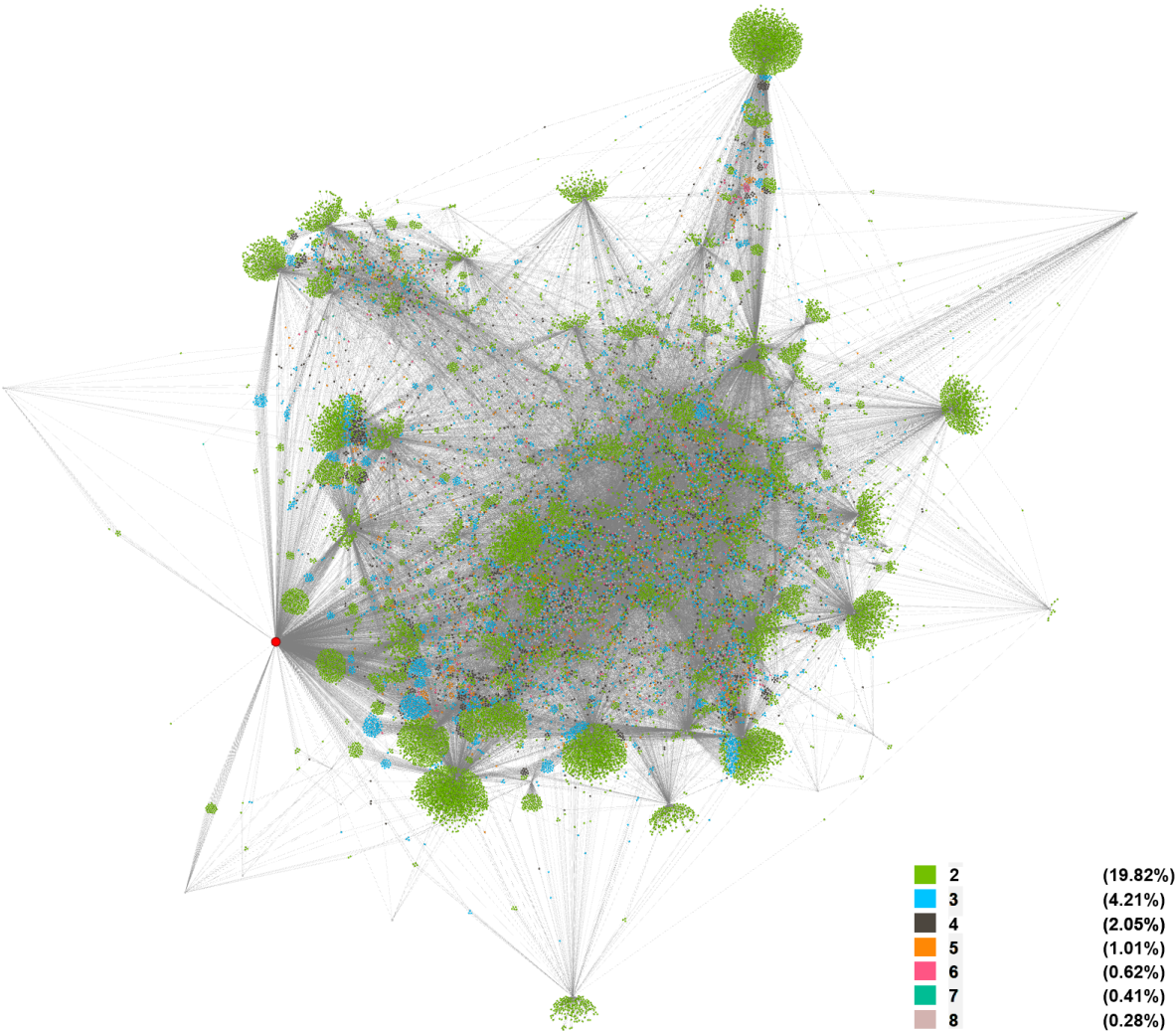
Centrality measures provide different perspectives on influence and importance within a network. By analyzing degree, betweenness, closeness, eigenvector centrality, and PageRank, we gain a more nuanced understanding of how nodes interact and which users hold key positions in the Bluesky network. Each measure captures a unique aspect of connectivity and prominence.

From our previous analysis we found that Gianis was the node with the highest degree, therefore I was expecting him to be the most influential node in the network. But other centrality measures show that that's not necessarily the case.



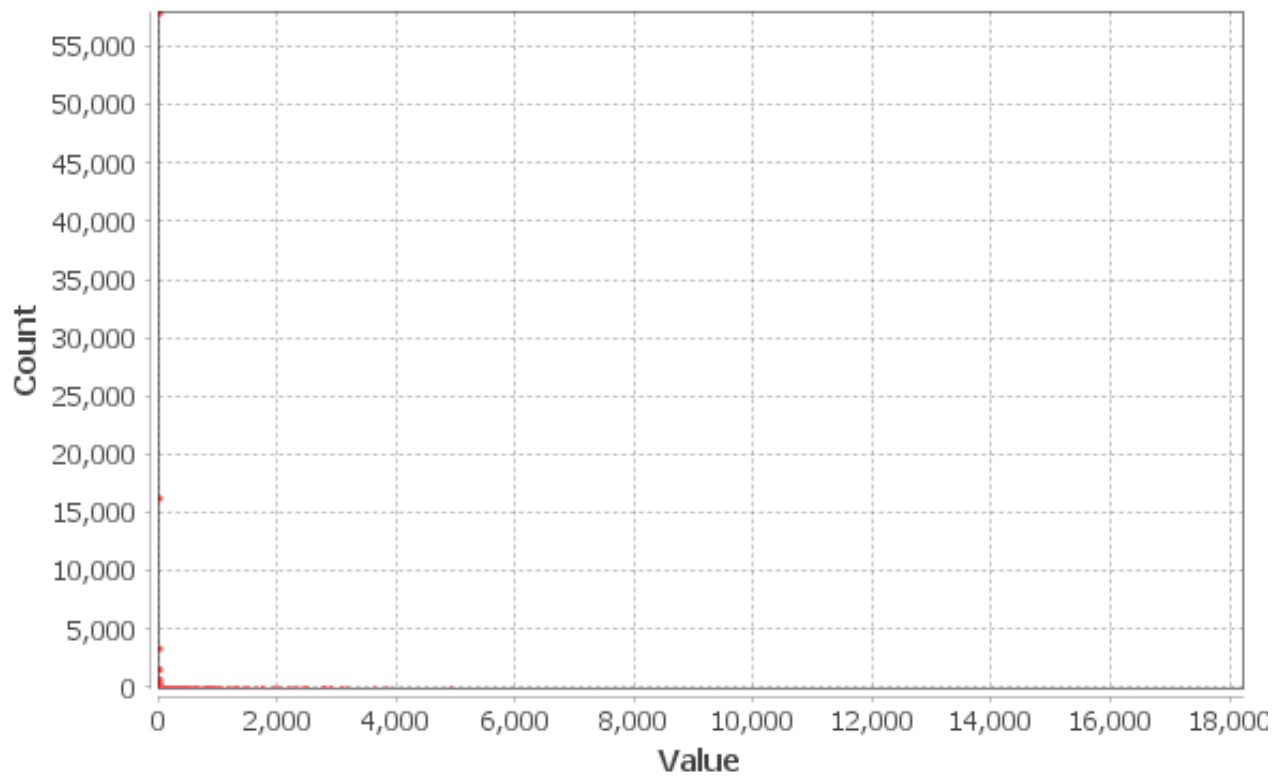


**Figure 3:** The Graph colored in respect to the Node Degree

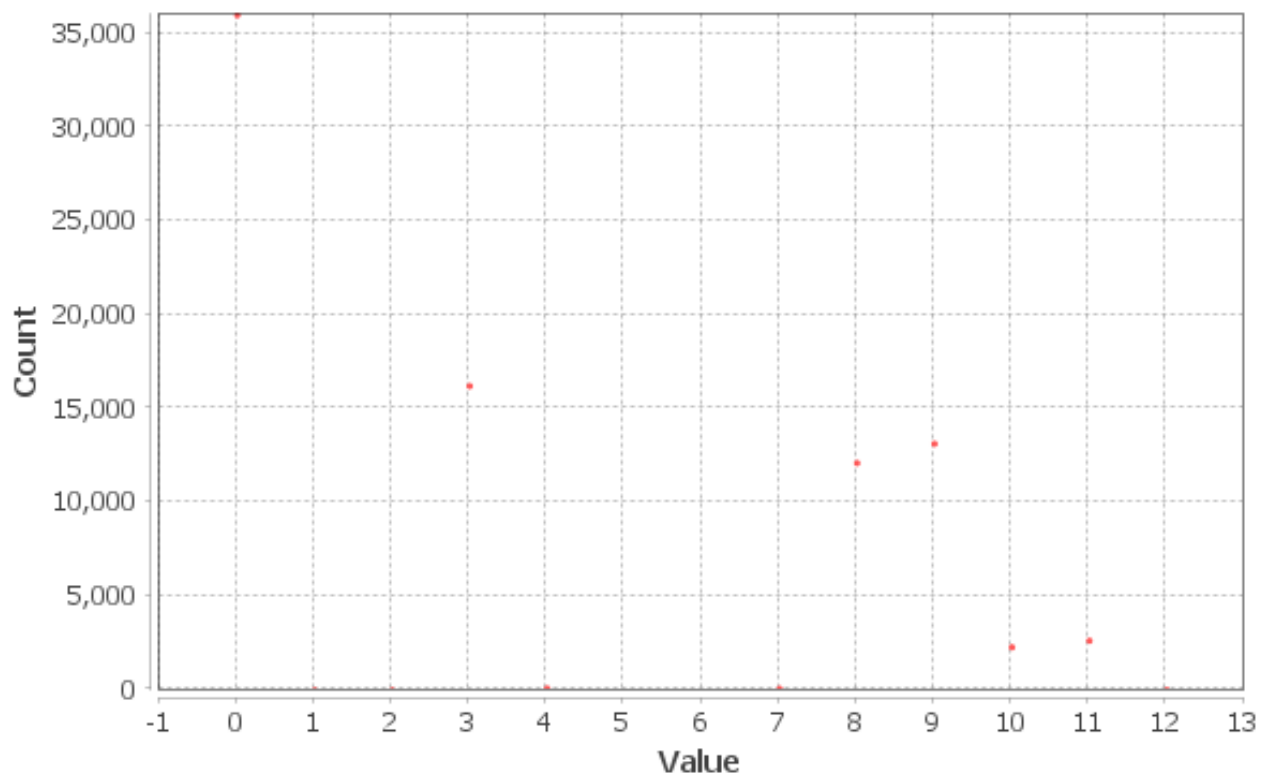


**Figure 4:** The Graph colored in respect to the Node Degree excluding nodes with a degree of one

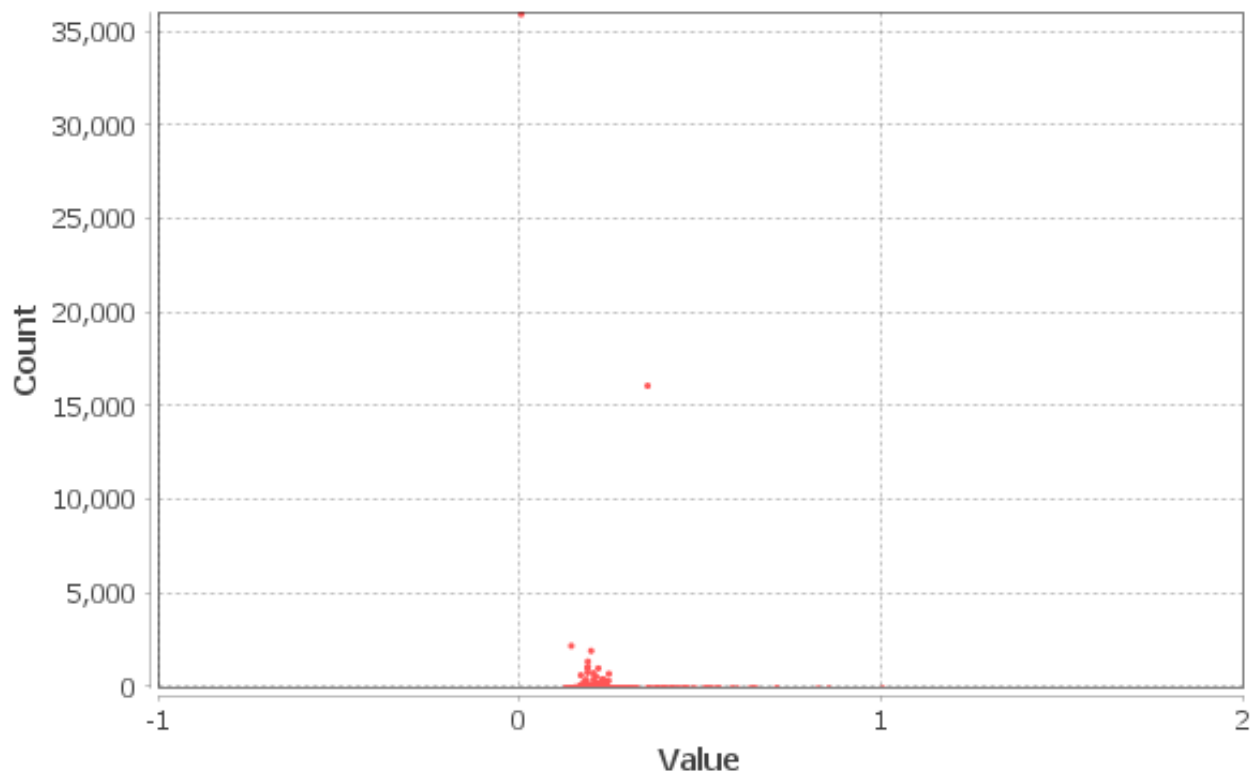




**Figure 5:** *The Distribution of node degree*



**Figure 6:** *The Distribution of Eccentricity*



**Figure 7:** *Closeness Centrality Distribution*

For closeness centrality the number one account was @atproto.com, this is the account for the AT Protocol Developers. The AT protocol (Authenticated Transfer Protocol, pronounced) as described in [10] is the decentralized and open social media protocol which Bluesky is built on. This is expected, as Bluesky is still quite new most people who use it are probably familiar with the technology and the protocol that makes it work, this makes them more inclined to follow the Developer Account making it a quite influential node. In this metric Gianis is 25th with a closeness centrality of 0,51. The closeness centrality distribution is found in figure 7

For betweenness centrality the number one account is @paulftremlett.bsky.social. Dr Paul-Francois Tremlett is a Senior Lecturer in Religious Studies at the Open University in Milton Keynes, England. He is the only account which is both in the top ten for the measures of in-Degree and out-Degree (He has around fourteen hundred followers

and is following nineteen hundred accounts). I believe this contributes the most to the high betweenness centrality; but I could not draw a definite conclusion. My other assumption, looking through his biography, was that his high involvement in different areas, such as literature, ethnographic research and teaching, make him an account with followers and following from completely different backgrounds and areas of interest.

Eigenvector centrality assigns scores based on the principle that a node is important if it is connected to other important nodes, calculated as the dominant eigenvector of the adjacency matrix. For eigenvector centrality let's start with account number four, which was again @paulftremlett.bsky.social with a measure of 0,425 who we talked about earlier. The number two account was Gianis with a measure of 0,572. Number one for eigenvector centrality is @ncecire.bsky.social who has a score of 1 (that's expected as eigenvector is relative). The big difference between them and the

second-place account shows that they seem to have quite a large number of connections to influential nodes in the network. This seems to correlate to the fact that this account has a large number of followers and followees which are all high-stake individuals. They, of course, follow all other the previously mentioned accounts which are collectively highly influential.

## 10 Clustering Effects

The average clustering coefficient quantifies the likelihood of a node's neighbors being connected. It is 0,019 which is low. This denotes that the nodes in the network don't have a high degree of local interconnectedness, and the network is relatively sparse in terms of local clustering. This is expected regarding our analysis up until now.

My clustering coefficient distribution is in figure 9, it is as expected left hump distributed.

There are 6.204 triangles in the network and triadic closure is very low, the fact that most of the nodes have a degree which is one or two is the first indication that the transitivity (the percentage of closed triads) will also be exceptionally low. Transitivity was calculated as 0000079 or, to put it plainly, one in about thirteen thousand triads are closed. Triadic closure is a phenomenon which is highly anticipated in social networks; therefore, I attribute the fact that transitivity is so low to the relative novelty of BlueSky at the time on analysis.

## 11 Bridges

Bridges are edges in a graph that, if removed, increase the number of connected components. Local bridges are edges whose removal increases the shortest path length between their endpoints. Earlier in the analysis, it was outlined that nodes degrees are

usually either one or two and triadic closure is low. This leads to most edges also being bridges, as removing them would create a disconnected component of a singular node. The number of bridges is 68.343 and the number of local bridges is 107.631. As stated earlier, my assumption is that the relative novelty of BlueSky is making subsets of its network not yet model what we expect from social networks (as in network science; not social media networks).

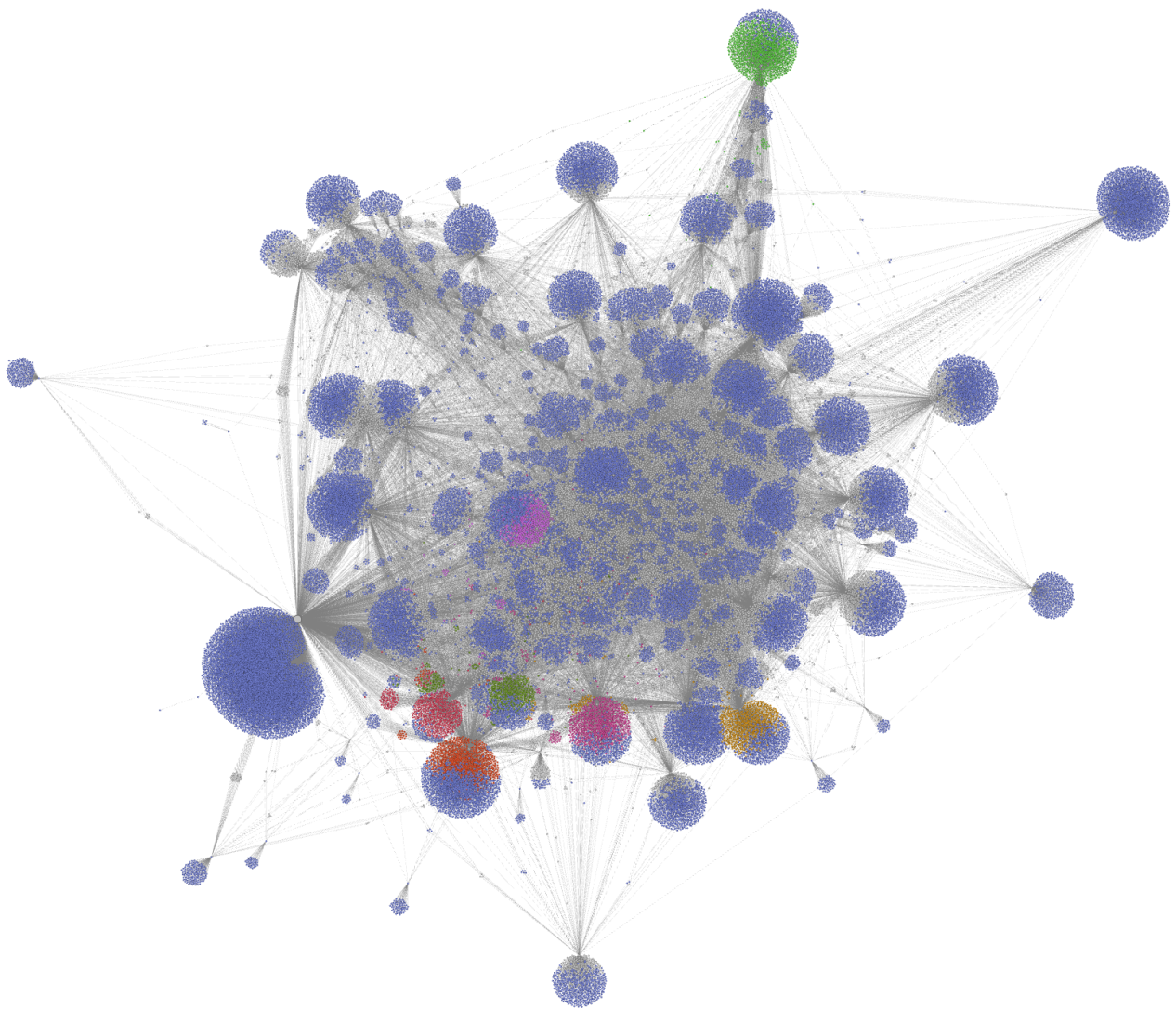
## 12 Gender and Homophily

We were taught that to establish whether a network is homophilous or not we have to measure the cross-gender edges and compare them to what we would expect if they were formed randomly. The problem is what do we define as "Gender" in our network. The first option would be to use actual assigned sex, so male and female, to categorize the nodes and then calculate are cross-gendered edges. Although this would be interesting, I don't think it would add value to our analysis and there is also no easy way to find this.

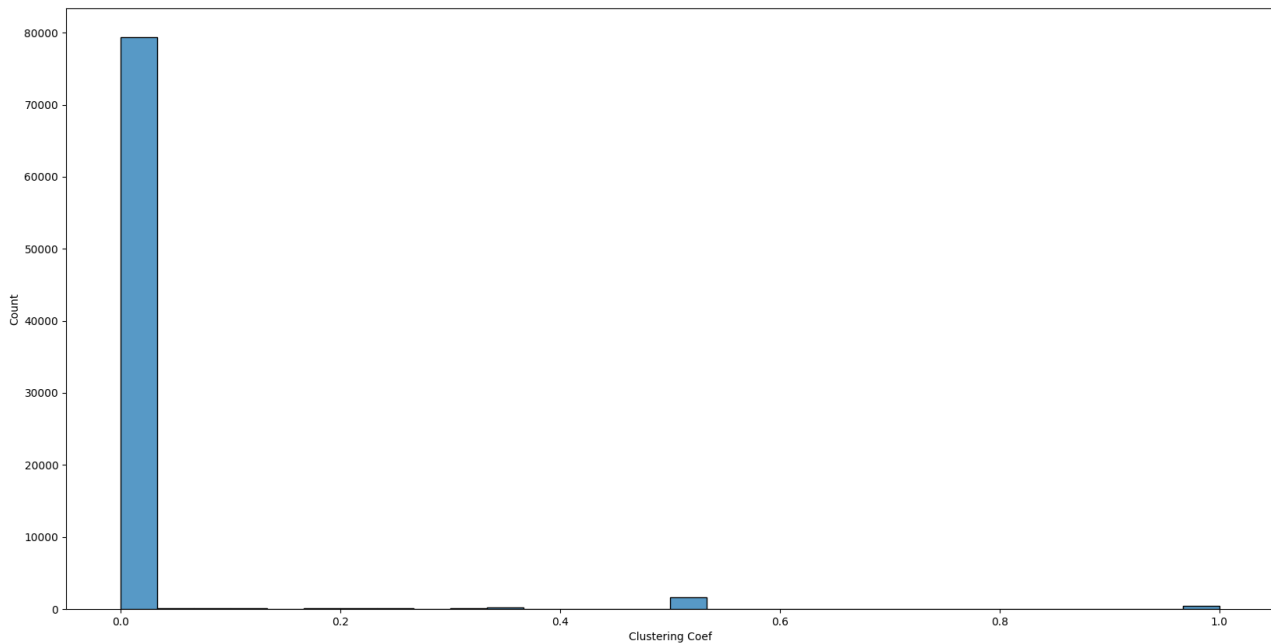
### 12.1 Political Gender

My analysis of Gender will work as follows. Looking at the Description of each profile we can try to determine if the person is politically "left leaning" or not. Then we can assign 1 to "left leaning" and 0 to "not left leaning" individuals. I say leaning as you can't be sure by using the Description of a profile. This "Gender" will help us detect if Yannis', who is a left politician, network is mostly comprised of left-leaning accounts and whether there is an expected homophily between these categories.

To determine if an account is left leaning, I looked for phrases that would indicate they are in the Description. Some examples are



**Figure 8:** *the Graph colored in respect to the Eigenvector Centrality*



**Figure 9:** *The Distribution of Clustering Coefficients*

pronouns, “Just stop oil”, “lesbian”, “trans”, “bi”, “union”. The full list of phrases is found in the appendix. These phrases were compiled after looking through a sample of descriptions by hand and using common judgement to affiliate them with a political ideology. For example, pronouns (even though they are a part of the English language) are usually only mentioned in profile descriptions by left leaning individuals, non-straight sexual orientations or proudness is also closely affiliated with the LGBTQ community. From [11] “Historically, Sex Has Been a compelling issue for the left”. Even though this method is not perfect it’s the easiest way to garner a (obviously partly flawed) sense of political affiliation for our nodes.

I calculated the Assortativity Coefficient, the Homophily Index and both the expected and actual amount of cross gendered edges. The Assortativity Coefficient is a measure of the tendency of nodes in a network to connect to other nodes that are like themselves. It’s used to quantify whether individuals are likely to associate with others who have similar characteristics [12]. The homophily

index, a concept from sociology, measures the extent to which individuals are likely to form connections with others who are similar in some way (such as age, race, interests, etc.). In our analysis similarity is political affiliation. The Assortativity Coefficient is 0,0560 and indicates a very weak assortative behavior, meaning the network shows slight preference for similar connections, but it’s not strong. The Homophily Index is 0,6464 which suggests moderate homophily. Keep in mind that the classes are imbalanced therefore the Assortativity Coefficient is more correct.

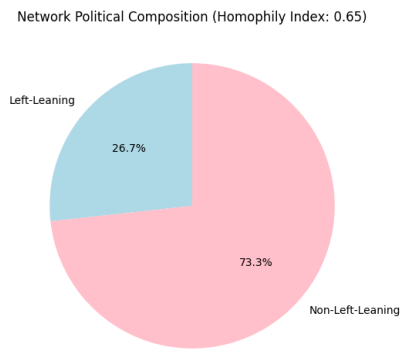
This is also backed up by the difference between the actual amount of cross-gender edges being 47.070 edges and the expected amount of cross-gender edges being 52.145 edges. As the expected amount is larger than the actual amount there is slight evidence of Homophily in the network.

The expected amount of homophily is given by the following equation:

$$\text{Expected Amount} = 2pq |E|$$

where  $p$  is the proportion of one type of characteristic in a population (in our case left-leaning),  $q$  is the proportion of the other





**Figure 10:** The networks political composition

type of characteristic in the population (in our case not left leaning) and  $|E|$  the amount of the nodes (in our case labeled).

## 13 Graph density

Graph density shows us how close a network is to the maximum possible amount of edges. The maximum number of edges is found using:

$$\max = \frac{n(n-1)}{2}$$

For our network that will equal 3.5 billion edges. It's obvious that our network is sparse mirroring most social and computer networks [13]

$$\max = \frac{82733 \times 82732}{2} = 3.422.333.278$$

## 14 Community

Running the Modularity Community Detection, I found a modularity of 0,736 and 26 Communities with their size distributed as in figure 11. Also, the preliminary inference I made earlier seems to be correct, by far the largest Class is class number two (Colored with blue) which is the left non interconnected component of followers.

As we know, a network can have high modularity (clear community structure) and low triadic closure when the communities are tightly knit but only sparsely connected to other communities, which prevents much triadic closure from happening across different subgroups. As extensively emphasized, qualitatively and now quantitatively, this is the case with our network.

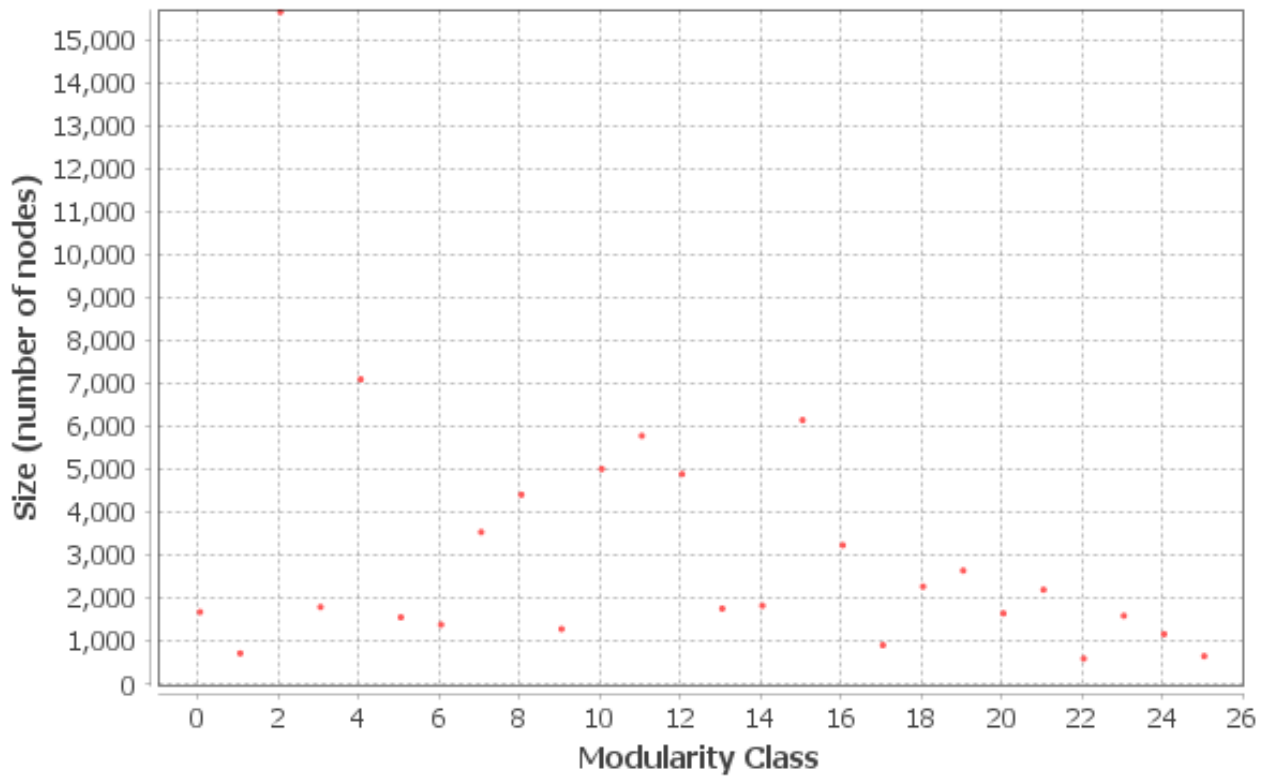
## 14.1 Interesting Communities

After careful analysis of the remaining communities, as I had previously already analyzed community number two, no other interesting patterns between them became evident.

## 15 PageRank

PageRank is an algorithm designed to measure the influence ranking of nodes within a network. It operates on the principle that a link from one node to another acts as a vote of confidence, and importantly, that votes from influential nodes carry more weight. Essentially, a node's PageRank score reflects the quantity and quality of its incoming links. The application of PageRank in our network was conducted using the included Gephi implementation. The results are found in figure 12. With PageRank we seem to have come full circle. Yanis is once again at the top with a score of 0.078 and is followed closely by second place @atproto.com with a score of 0.066 whom we saw dominate the closeness centrality. Well behind coming in third is @nce-cire.bsky.social with a score of 0.032, remember they were first in regards to eigenvector centrality.

In conclusion, PageRank analysis confirms that Yanis is most likely the most influential node in the network. This distinct ranking, when compared to closeness and eigenvector centralities, underscores that different



**Figure 11:** *The Distribution of our Communities*

centrality measures capture varied aspects of influence, providing a more comprehensive view of node importance within the network but not drawing a definitive conclusion.

## 16 Notae Auctorum

Throughout this Coursework I refer to Prof. Varoufakis by writing his first name with one n, this is something which diverges from the usual spelling of the name in Greek which is with a double n. In order to fulfill the right of every person to self-determination I have, in accordance with ISO 843, transcribed his preferred name in to the Latin alphabet, which gives us “Gianis”.

In multiple occasions I was unable to determine if some account’s use Male or Female pronouns, to keep things easy I used the singular “they” pronoun. This is because to refer to an unknown person, in English,

“they” typically occurs as an indeterminate antecedent [14]. This is not meant to push any sort of “Woke” agenda as some people tend to think, but it’s just how the English language works.

I pulled my nodes and edges very early in the semester when the Bluesky network was still getting bigger, therefore many of my conclusions might seem out of touch with the current and more interconnected Bluesky Networks. Unfortunately, I did not have the time to re-run the analysis after the boom caused by the US election.

It is important to note that the, automatically followed, Bluesky node (@bsky.app) was excluded from this analysis, as it artificially skewed the results and introduced distortions in the network measurements.

In Greek notation, the period is used as the thousand separator while the comma serves as the decimal separator. This is the reverse of the convention commonly used in English-speaking countries. In this course-

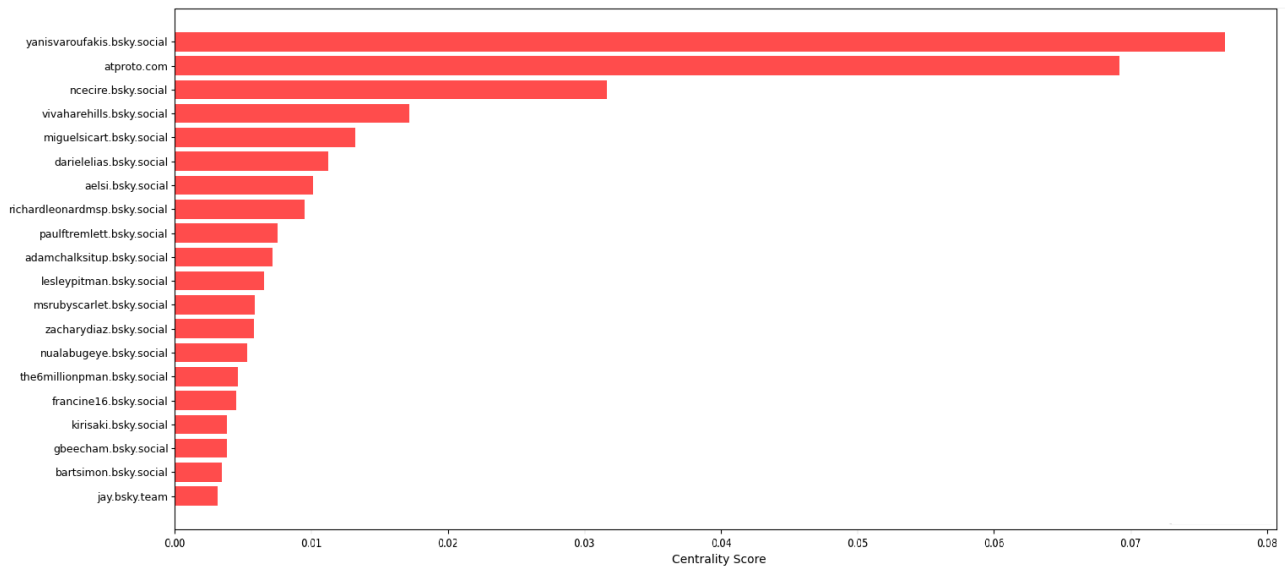


Figure 12: The top 20 account in regards to Page Rank

work I follow the Greek notation, the thousand separator is the period (.) and the decimal separator is the comma (,).

In regards to the question about giant components in the coursework: To be honest I found it weird, as it's impossible to not have one weakly connected giant component due to the nature of the network pull (The plugin finds nodes via edges).

## 17 Appendix

Keywords used to classify descriptions as left leaning are:

- pronouns
- just stop oil
- donate
- mental health
- communist
- nature
- vegan
- lefty
- queer
- free palestine
- lesbian
- trans
- bi
- union
- climate change
- activist
- feminist
- equal non\_lefts
- blm
- lgbt
- progressive
- democratic socialist
- they/them
- he/him
- she/them
- he/them
- she/her
- they/he
- liberal
- climate
- palestinians
- gaza
- antifascism
- socialist
- leninist
- anti imperialist
- universal healthcare
- chinese buddhism
- #freepalestine
- luigi did nothing wrong
- refugee
- anti-terf
- genocide

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## List of Figures

|   |                                                                                                |    |
|---|------------------------------------------------------------------------------------------------|----|
| 1 | The Graph colored in respect to the Harmonic Closeness of nodes . . . . .                      | 4  |
| 2 | The Harmonic Closeness Centrality Distribution . . . . .                                       | 5  |
| 3 | The Graph colored in respect to the Node Degree . . . . .                                      | 7  |
| 4 | The Graph colored in respect to the Node Degree excluding nodes with a degree of one . . . . . | 8  |
| 5 | The Distribution of node degree . . . . .                                                      | 9  |
| 6 | The Distribution of Eccentricity . . . . .                                                     | 9  |
| 7 | Closeness Centrality Distribution . . . . .                                                    | 10 |
| 8 | the Graph colored in respect to the Eigenvector Centrality . . . . .                           | 12 |
| 9 | The Distribution of Clustering Coefficients . . . . .                                          | 13 |

10 The networks political composition . . . . . 14

11 The Distribution of our Communities . . . . . 15

12 The top 20 account in regards to Page Rank . . . . . 16